Latest Technologies to Enhance the Performance of Memories

Several initiatives have been taken to improve the performance of memories. The aim is to achieve lower power consumption by memories and higher memory speed. Hybrid DRAM, Flash and CPU module architecture speed up in-memory operations significantly.

We often come across problems with the hardware of our phones, tablets and laptops. These problems could be related to storage or the time it takes to access information. One of the new play areas of smartphones is artificial intelligence (AI), which requires faster and more memories.

3D XPoint technology is a new class of non-volatile memory (NVM) that can help turn immense amounts of data into valuable information in real time. It has about 1000 times lower latency and exponentially greater endurance than NAND. It can deliver game-changing performance for transactional workloads and big data applications.

Due to data explosion, 3D memories technology is a big innovation. Cross-point architecture means that there are vertically-placed conductors that can connect 128 billion densely-packed memory cells in a very short space. This compact structure results in an astoundingly high level of performance and density. New technologies such as JBOF (just a bunch of flash) and JBOD (just a bunch of disks) are a boon for enterprise-level usage, especially in data centres, where more efficient flash and drives are required.

Manufacturing memories for improved performance

Sambit Sengupta, associate director - field applications, Avnet India, says, "For some mission-critical applications, field programmable gate arrays (FPGAs) can be used to leverage new technologies such as an automated processor, where memory forms the basis of the processing engine. So, the role of a co-processor along with an FPGA can be utilised efficiently. The memory sits inside the dual in-line memory module (DIMM), making the memory integrated with the co-processor.

"In the future, most applications will require sensors and embedded processors to be in always-on mode, or be able to go live in a very short time to take care of power requirements. Such applications will require high-speed serial NOR flash. We support customer designs with partners like International Space Science Institute and Micron for these technologies. Many new consumer applications and daily-use technology enablers already use low power double data rate (LPDDR) devices, embedded multimedia controllers (eMMCs) and multi-chip package (MCP) memories."

Redundant array of independent disks (RAID): Here, when you connect two or three drives in the system simultaneously, together these act as one large-volume fast drive. RAID is set up as one system drive and duplicates the data for real-time backup.

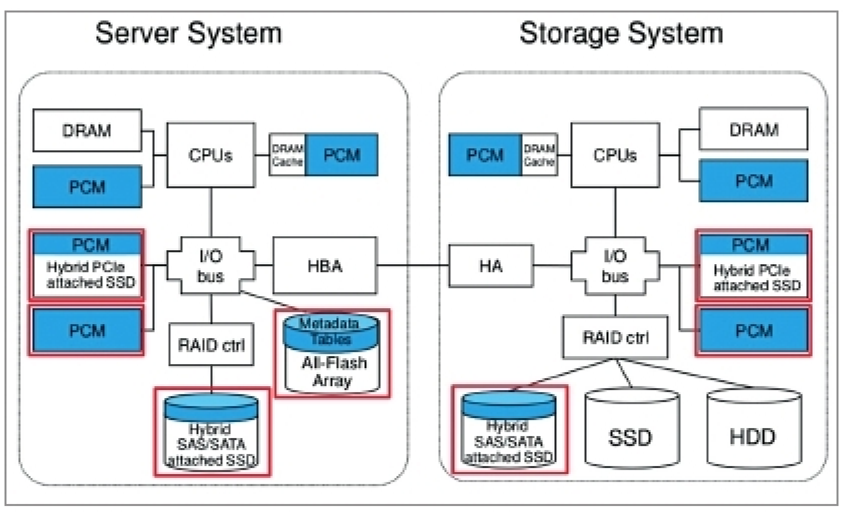

Figure 1: Computation and storage memory (Credit: www.extremetech.com)

DIMM: DIMM is an emerging memory technology that uses NAND flash chips (NVM) in standard cards to directly connect to memory slots. This solves the problem of power loss, but also reduces system performance. Flash is slower at reading and writing data than DRAM.

Dynamic RAM (DRAM)-based system memory, used for active computing, loses its data when the power goes away. As the use of in-memory database systems (to keep data in active memory) increases for rapid processing, developers must ensure that data is retained even after power loss.

Phase-change memory (PCM): This changes the phase of a special kind of glass (Ge2Sb2Te5) within the bit cell. It heats up when a programming current flows through the cell. A higher heating current that is removed early causes the glass to solidify into an amorphous, non-conductive state. Slow heating at a lower initial temperature solidifies the glass in a conductive crystalline structure. Writing of a bit from a one to zero of a PCM product, such as 3D XPoint, is handled by electronically flipping the resistance of the individual cell. This makes PCM much faster than NAND flash, and it is non-volatile like flash.

3D XPoint. The revolutionary new-generation memory technology, 3D XPoint is a solid-state drive (SSD) that provides high speed in accessing memory. It is a combination of RAM and flash storage, so it is four times denser than traditional RAM. It is a non-volatile form of memory. 3D